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Q.1 Explain the Computer organization and architecture with examples. Draw the Von Neumann's architecture of Computers. Explain the Working of various CPU registers.

→ Computer architecture refers to those parameters of a Computer system that are visible to a programmer or those parameters that have a direct impact on the logical execution of a program.

Examples:- a) Instruction set designing

b) Instruction format

c) No of bits used to represent various types of data

d) Different addressing mechanism to access data.

Computer architecture refers to those parameters of a Computer system that are visible to a programmer or those parameters that have a direct impact on the logical execution of a program.

Computer organization:- it refers to the operational units and their interconnections that realize the architectural specifications. Examples of organizational attributes include those hardware details transparent to the programmer, such as Control signals.

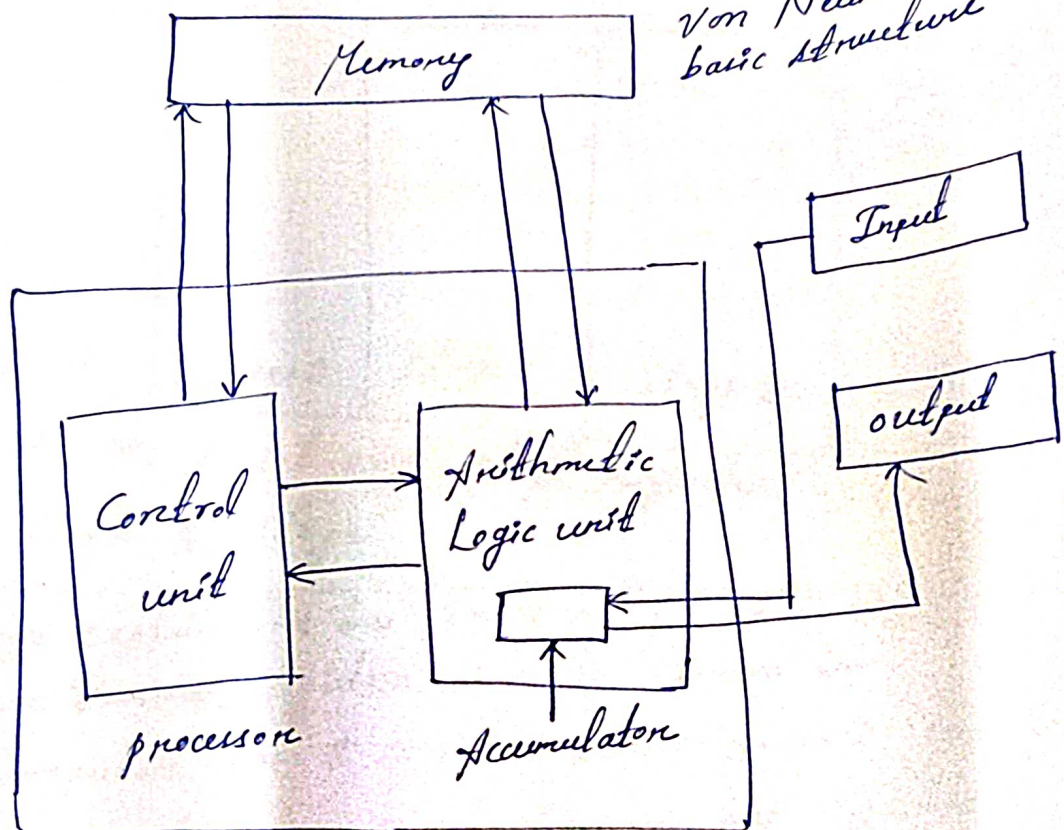


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→ Von Neumann's Model :->

This architecture refers to a design model for computers where the processing unit, memory, and input-output devices are interconnected through a single, central system bus. This architecture was first proposed by John Von Neumann.



1.] serial In - serial out (SISO) shift Register :->

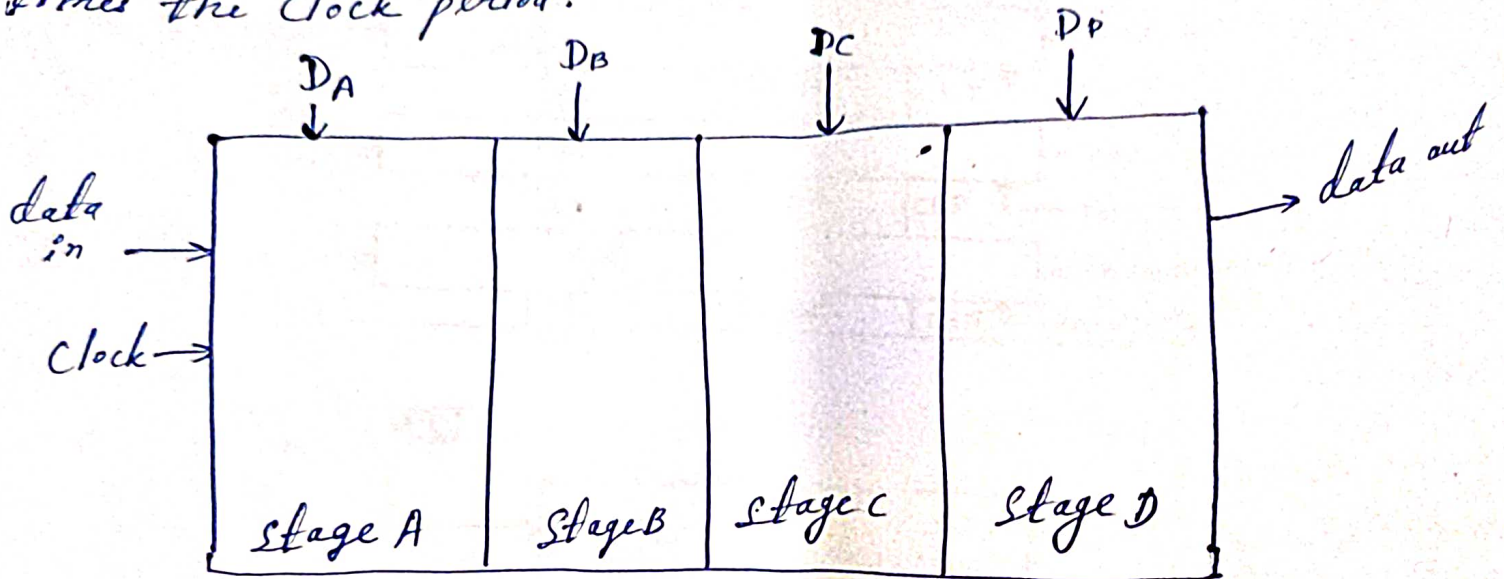
In the serial in - serial out shift register, data is input serially until it reaches the output. At this time, it exists in a serial out shift register.



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- Parallel-in/Serial-out shift register $\rightarrow$ The parallel-in/serial-out shift register stores data, shifts it on a clock by clock basis, and delays it by the number of stages times the clock period.



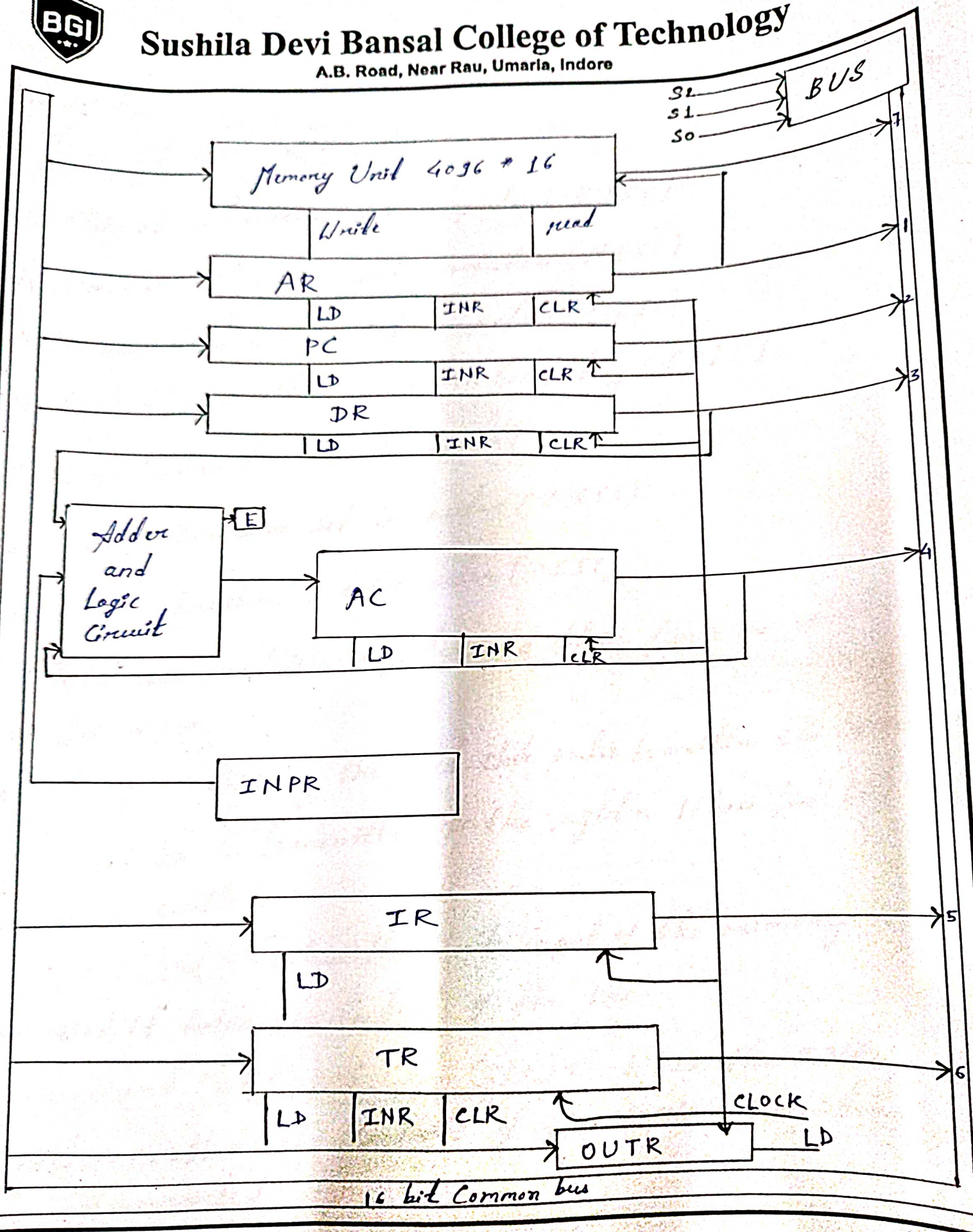
parallel-in, serial out shift register
With 4-stages.

Q.2 Draw and explain the bus structure for that data transfer between various registers and the Common bus.
 $\rightarrow$ A basic Computer has 8 registers, memory unit and a Control unit. The diagram of the Common bus system is as shown below.



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is used.

Types of instruction Set :- $\rightarrow$

1. Reduced instruction Set Computer (RISC)
2. Minimal instruction set Computers (MISC)
3. Complex instruction set Computer (CISC)
4. Explicitly parallel instruction Computing (EPIIC)
5. Very ^{long} instruction Word (VLIW)
6. Zero instruction set Computer (ZISC)
7. One instruction set Computer (OISC)

There are multiple registers and their functions which are as follows :- $\rightarrow$

• Load (LD) :- $\rightarrow$ During the next clock pulse transition the information from the bus is transmitted to the register whose Load (LD) input is enabled.

• Memory Unit :- $\rightarrow$ When the Write input of the memory is activated, it holds the Content of the bus.

• Increment (INR) and Clear (CLR) :- $\rightarrow$ When the INR signal is enabled, the Contents of the specified register are incremented



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- Address Register (AR) :- The address of the memory for the next read and Write operation is determined.
- Program Counter (PC) :- The address of the next instruction that is to be read from the memory is saved.
- Data Register (DR) :- The data register includes the data to be written into memory or data that is to be read from the memory.
- Accumulator (AC) :- Accumulators are beneficial in executing the register micro-operations including Complement, shift, etc.
- Instruction Register (IR) :- The IR stores the copy of the instruction that the processor has to implement.
- Temporary Register (TR) :- The temporary storage for variables or results is supported by the temporary register.
- Input Register (INPR) :- It includes 8 bits to hold the alphanumeric input information.
- Output Register (OUTPR) :- The data is received from AC and moved to the output device.

Q. 3 Define the instruction format. Discuss instruction types. Explain the I/O system in detail. Also, explain the Instruction Cycle.

→ instruction formats refer to the way instructions are encoded and represented in machine language.



There are several types of instruction formats, including Zero, one, two, and three-address instructions. Each type of instruction format has its own advantages and disadvantages in terms of Code size, execution time, and flexibility. Modern Computer architectures typically use a combination of these formats to provide a balance between simplicity and power.

Types of instructions: →

- 1] Arithmetic Instructions.
- 2] Branch Instructions.
- 3] Data Transfer Instructions
- 4] Logic Instructions
- 5] Bit-oriented Instructions.

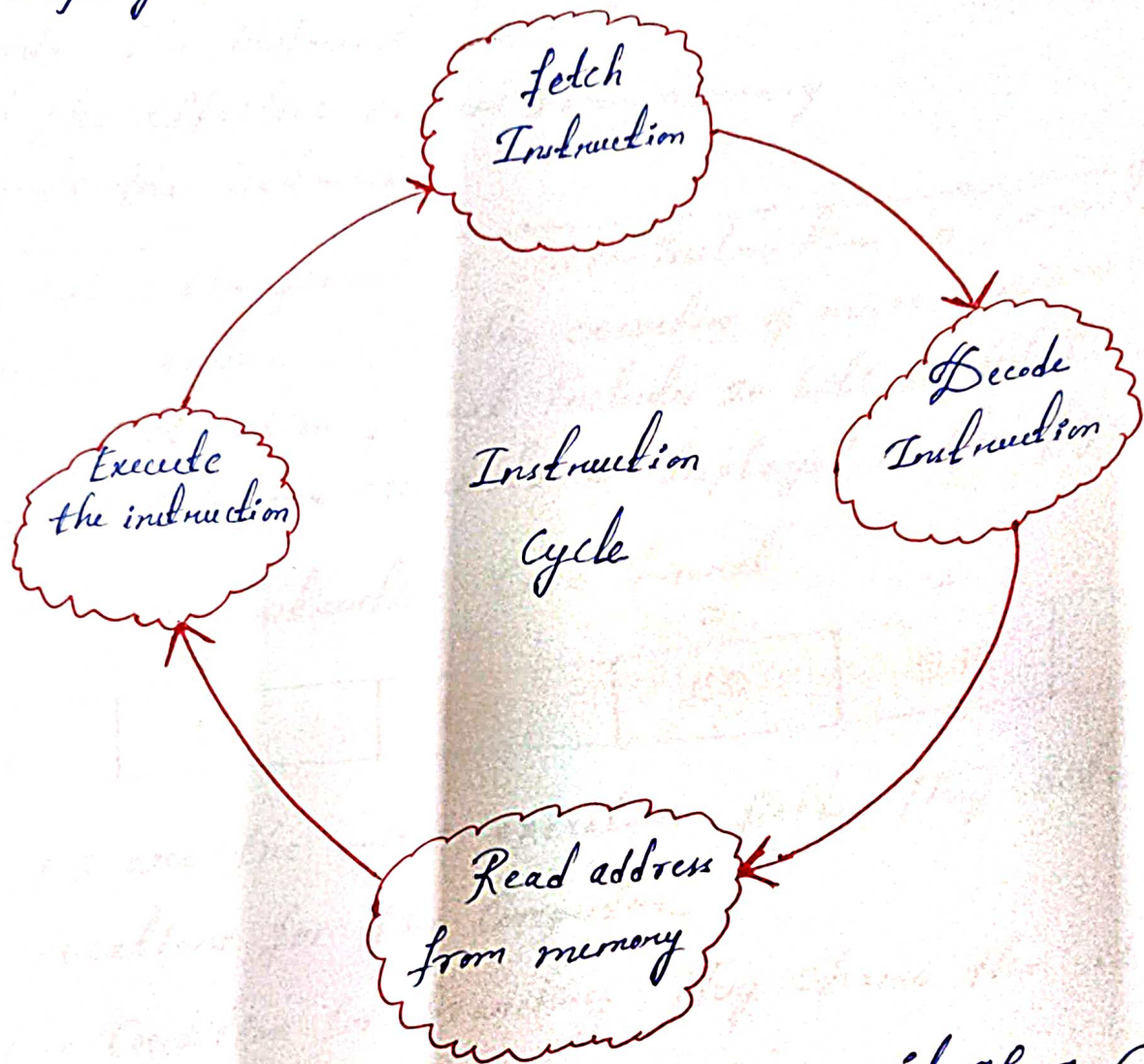
Within the World of I/O, there are two primary types of devices; input devices and output devices. Input devices only allow for input of data to another device. An example of an input device is a mouse, as it can only receive and display the data that is sent (output) from a Computer. output devices only receive the output of data from another device. An example of an output device is a monitor, as it can only receive



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and display the data that is sent from a Computer



A program residing in the memory unit of a Computer consists of a sequence of instructions. These instructions are executed by the processor by going through a cycle for each instruction.

In a basic Computer, each instruction cycle consists of the following phases: →

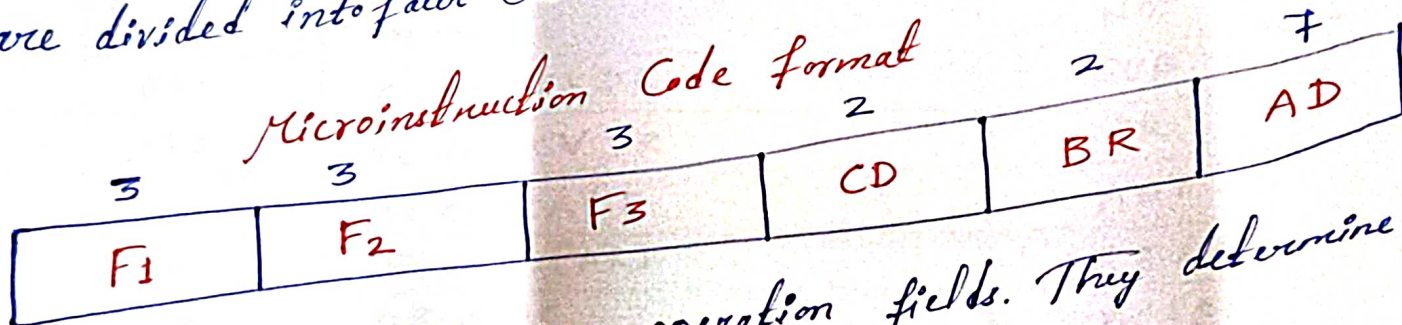


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1. fetch instruction from memory
2. Decode the instruction
3. Read the effective address from memory.
4. Execute the instruction.

Q. 4 What is the format of micro-instruction in Computer Architecture explain. Explain the execution of micro-instruction.
→ A microinstruction format includes 20 bits in total. They are divided into four elements as displayed in the figure.



F₁, F₂, F₃ are the micro-operation fields. They determine micro-operations for the Computer.
CD is the Condition for branching. They choose the status bit conditions.

BR is the branch field. It determines the type of branch.
AD is the address field. It includes the address field whose length is 7 bits.

The micro-operations are divided into three fields of three bits each. These three bits can define seven different micro-operations. In total there are 21 operations as displayed in



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the table.

	Name:	Code	Symbol
F ₁	000	None	NOP
	001	$AC \leftarrow AC + DR$	ADD
	010	$AC \leftarrow 0$	CLRAC
	011	$AC \leftarrow AC + 1$	INCA
	100	$AC \leftarrow DR$	DRTAC
	101	$AR \leftarrow DR (0-10)$	DRTAR
	110	$AR \leftarrow PC$	PCTAR
	111	$AC \leftarrow AC + DR$	WRITE
F ₂	000	None	NOP
	001	$AC \leftarrow AC + DR$	SUB
	010	$AC \leftarrow AC \vee DR$	OR
	011	$AC \leftarrow AC \wedge DR$	AND
	100	$DR \leftarrow M[AR]$	READ
	101	$DR \leftarrow AC$	ACTDR
	110	$DR \leftarrow DR + 1$	INCDR
	111	$DR (0-10) \leftarrow PC$	PCTDR
F ₃	000	None	NOP
	001	$AC \leftarrow AC \oplus DR$	XOR
	010	$AC \leftarrow AC'$	COM
	011	$AC \leftarrow shl AC$	SHL
	100	$AC \leftarrow shr AC$	SHR
	101	$PC \leftarrow PC + 1$	INCP
	110	$PC \leftarrow AR$	ARTPC
	111	$DR (0-10) \leftarrow PC$	Reserved



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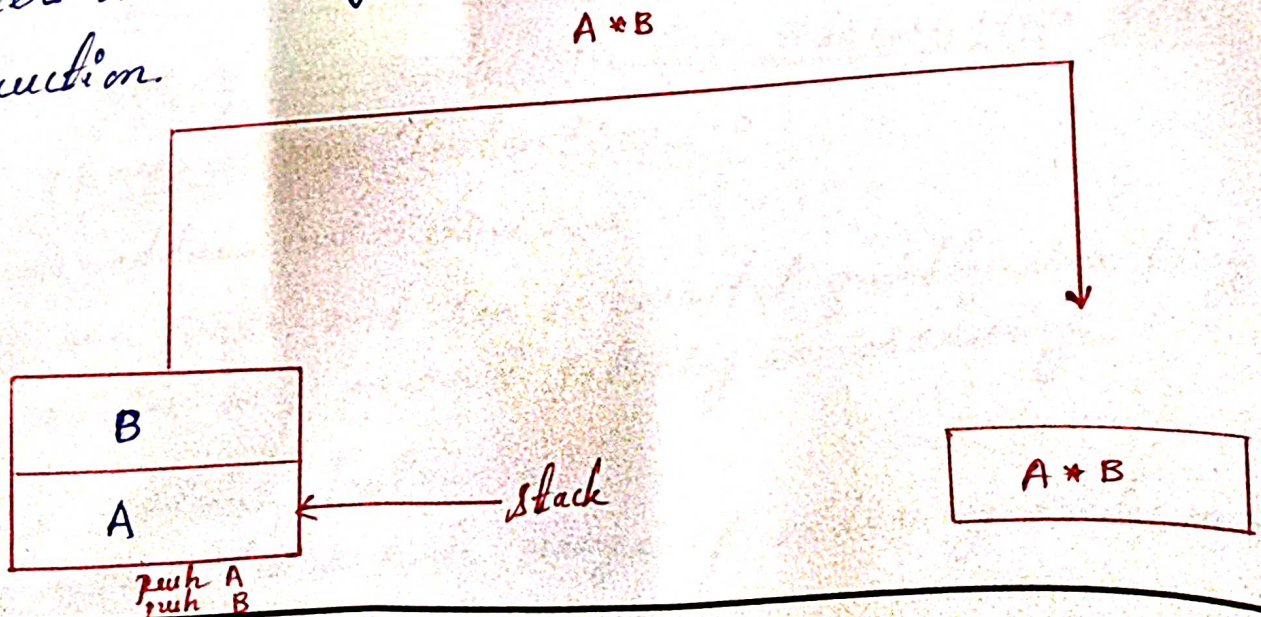
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Each microinstruction can have only three micro-operations, one from each field. If it uses less than three, it will result in more than one operation using the no operation binary code.

Q.5 Explain the Zero, one, two and three address instructions with examples. What are the different categories of the 8085-instruction set? Define the stack organization.

→ In Computer organization, instruction formats refer to the way instructions are encoded and represented in machine language. There are several types of instruction formats, including Zero, one, two and three-address instructions.

• Zero Address Instructions: → These instructions do not specify any operands. Instead, they operate on data stored in registers or memory locations implicitly defined by the instruction.





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- One address Instructions : → This uses an implied Accumulator register for data manipulation. One operand is in the accumulator and the other is in the register or memory location.

opcode	operand/address of operand	mode
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- Two Address Instructions : → These instructions specify two operands on addresses, which may be memory locations or registers.

- Three Address Instructions : → These instructions specify three operands on addresses, which may be memory locations or registers.

opcode	Destination address	source address	source address	mode
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- Advantages of Zero-Address, One-Address, Two Address and three-address Instructions : →

→ Zero-address instructions : →

They are simple and can be executed quickly since they do not require any operand fetching or addressing.

→ one-address instructions : →

They allow for a wide range of addressing modes, making



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them more flexible than zero-address instructions.

→ Two-address instructions: →

They allow for more complex operations and can be more efficient than one-address instructions since they allow for two operands to be processed in a single instruction.

→ Three-address instructions: →

They allow for even more complex operations and can be more efficient than two-address instructions since they allow for three operands to be processed in a single instruction.

Disadvantages of Zero-Address, One-Address, Two-Address and Three-Address Instructions: →

Zero-address instructions: →

They can be limited in their functionality and do not allow for much flexibility in terms of addressing modes or operand types.

One-address instructions: →

They can be slower to execute since they require operand fetching and addressing.

Two-address instructions: →

They require more memory space than one-address instructions and can be slower to execute since they require operand fetching and addressing.



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• Three - address instructions :-

They require even more memory space than two - address instructions and can be slower to execute since they require Operand fetching and addressing.

Q.6 What is the Register Transfer Language (RTL)?
Give some suitable examples. Explain the following micro - operation and draw a suitable block diagram for it.

$P: R2 \leftarrow R1$

→ "Register Transfer level" Code is a smaller subset of the full range of HDL Code. RTL describes Circuits at a level similar to the design description on a Schematic: flip-flops activated by fully - specified clocks, and Combinatorial logic between the flip-flops.

An example is the generation of an intermediate file format produced by a Compiler such as gcc, having the translation of C Code to machine language for a specific microprocessor. RTL may be conceived as a pseudo - Code description of an instruction set architecture, describing the dataflow beyond the data. following are some commonly used register transfer example with an example:



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- (1) Accumulator $\rightarrow$ This is the most commonly used register for storing data read from memory.
- (2) General-purpose Registers $\rightarrow$ These are used to store data on intermediate outcomes during the execution of a programme. Assembly programming is required to access it.
- (3) Special purpose Registers $\rightarrow$ Users do not have access to the special purpose Registers. These are Computer System registers.
- MAR $\rightarrow$ Memory Address Registers are the registers that store the memory unit's address.
 - MBR $\rightarrow$ This register stores instructions and data received from and sent from the memory.
 - PC $\rightarrow$ program Counter indicates the next Command to be executed.
 - IR $\rightarrow$ Instruction Register stores the to-be-executed instruction.
 - Register Transfer $\rightarrow$ The replacement Operator designates the information moved from one register to another in symbolic form.

$$R_2 \leftarrow R_1$$

It denotes the data transfer from register R_1 to register R_2 .
In most cases, we want the transfer to happen only under



specific Control Conditions. The following if-then sentence demonstrates this: If $(P=1)$, $(R2 \rightarrow R1)$.

The Control signal P is generated in the Control portion.

• Micro Operations $\Rightarrow$ Micro-Operations are Operations performed on data stored in registers. A micro-Operation is a simple operation that is carried out on data contained in one or more registers.

Example $\Rightarrow$ Load, Shift, Count, and Clear.

• Type of Micro-Operations $\Rightarrow$ The following are the different types of micro-Operations:

- 1] Micro-Operations that move binary data from one register to another are known as register transfers.
- 2] In registers, arithmetic micro-operations operate on numeric data stored.
- 3] Bit manipulation operations on non-numeric data are performed by logic micro-operations.
- 4] Shift micro-operations are data-based shift micro-operations.

Q. 7 Write and explain various addressing modes of basic Computers. What are the different types of addressing modes supported by 8085 microprocessors?



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→ The addressing modes in 8085 microprocessors are instructions used to transfer data and perform operations on them. An 8085 microprocessor uses five addressing modes: Immediate addressing mode, Register addressing mode, Register indirect addressing mode, Direct addressing mode, and Implicit addressing mode.

• Immediate addressing mode → the data on which the operation is to be performed is mentioned in the instruction. Instead of specifying an address, the instruction specifies an Operand along with the operation that is to be performed. The instruction is 2 bytes when there is 8-bit data and 4 bytes when there is 16-bit data.

Examples → MVI B 05, JMP address, LXI H 3000

• Register addressing mode → the instruction mentions a register which stores some data. The operation specified by the instruction is performed in the register specified by it. The instruction can also specify two registers.

Examples → ADD B, INR B, Move B, D, PCHL

• Direct addressing mode → the instruction specifies some address which stores the data. This memory location is



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mentioned in the instruction as an operand. The size of an instruction in the direct addressing mode is 3 bytes.

Examples $\rightarrow$ IN 35, LHLD address, LDA 2050 etc.

• Register Indirect addressing mode $\rightarrow$ the data on which the operation is to be performed is stored in some memory location. This memory location is specified in a register pair.

Examples $\rightarrow$ LDAX B, LXIH 9500, MOV A, M

• Implied / Implicit addressing mode $\rightarrow$ the operand is described implicitly in the definition of the instruction. The operand is specified with the opcode of the instruction.

Examples $\rightarrow$ RRC, RLC, CMA

Q. 8. What is the function of the Control unit? Differentiate Hardwired and Micro-programmed Control units?

$\rightarrow$ The main function of the Control unit is to fetch and execute instructions from the memory of a computer. It receives the input instruction/information from the user and converts it into control signals, which are then given to the CPU for further execution. It is included as a part of Von Neumann architecture developed by John Neumann. It is responsible for providing the timing signals, and control signals and directs the execution of a program by the CPU.



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Hardwired Control Unit

- 1] The hardwired Control unit induces the Control signals required for the processor.
- 2] Hardwired Control unit is quicker than a microprogrammed Control unit.
- 3] It is hard to modify.
- 4] It is more expensive as compared to the microprogrammed Control unit.
- 5] It faces difficulty in managing the complex instructions because the design of the circuit is also complex.
- 6] It can use limited instructions.

Microprogrammed Control unit

- 1] The microprogrammed Control unit induces the Control signals through microinstructions.
- 2] Microprogrammed Control unit is slower than a hardwired Control unit.
- 3] It is easy to modify.
- 4] It is affordable as compared to the hardwired Control unit.
- 5] It can easily manage complex instructions.
- 6] It can generate Control signals for many instructions.

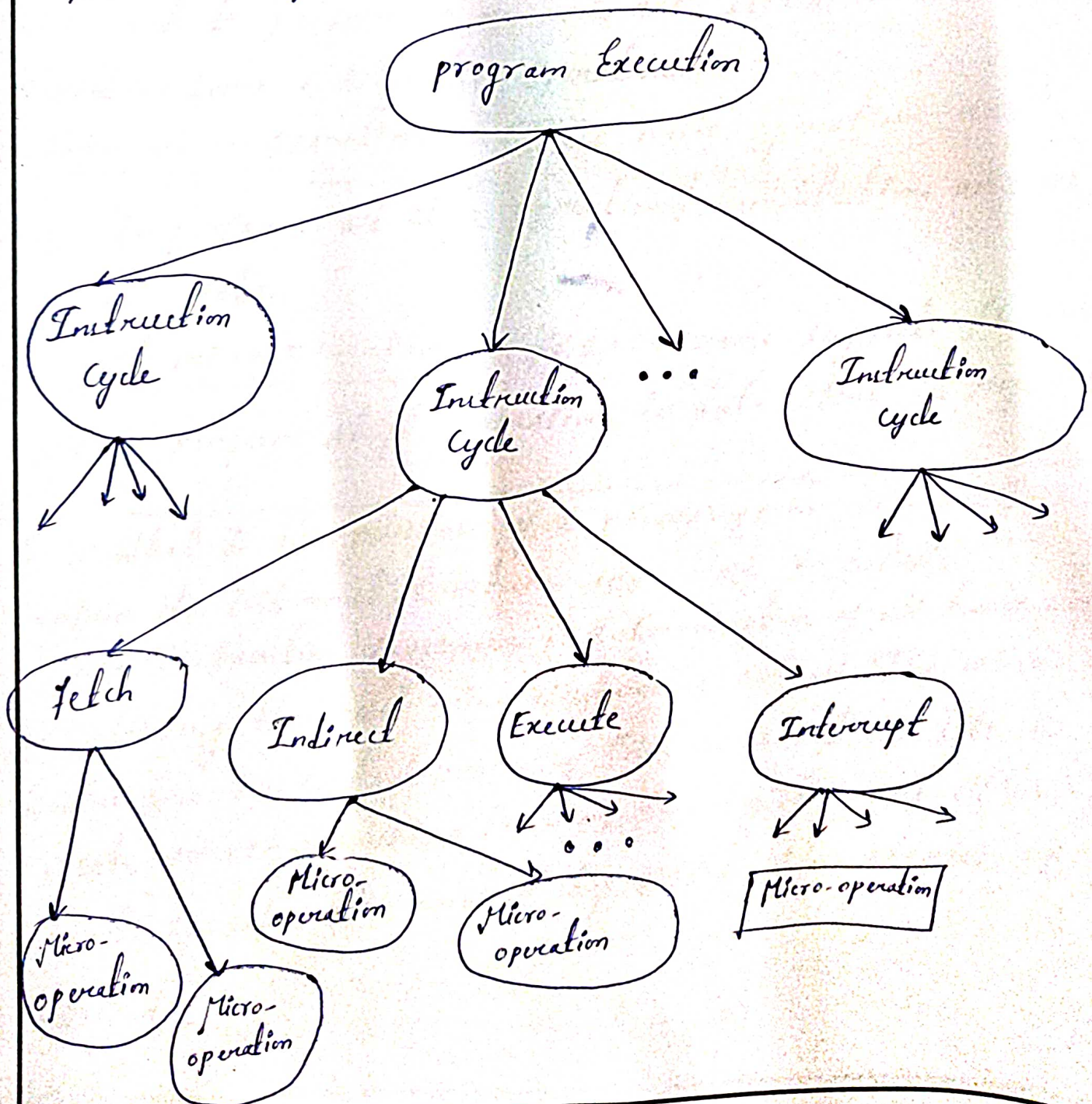


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Q.9 What do you understand by micro-operation? List types of micro-operations and explain them.

→ In Computer Central processing units, micro-Operations are detailed low-level instructions used in some designs to implement Complex machine instructions.





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- Register transfer micro-operations : $\rightarrow$ These type of micro-operations are used to transfer from one register to another binary information.
- Arithmetic micro-operations : $\rightarrow$ These micro-operations are used to perform on numeric data stored in the registers some arithmetic operations.
- Logic micro-operations : $\rightarrow$ These micro-operations are used to perform bit style operations/manipulations on non numeric data.
- Shift micro-operations : $\rightarrow$ As their name suggests they are used to perform shift operations in data store in registers.

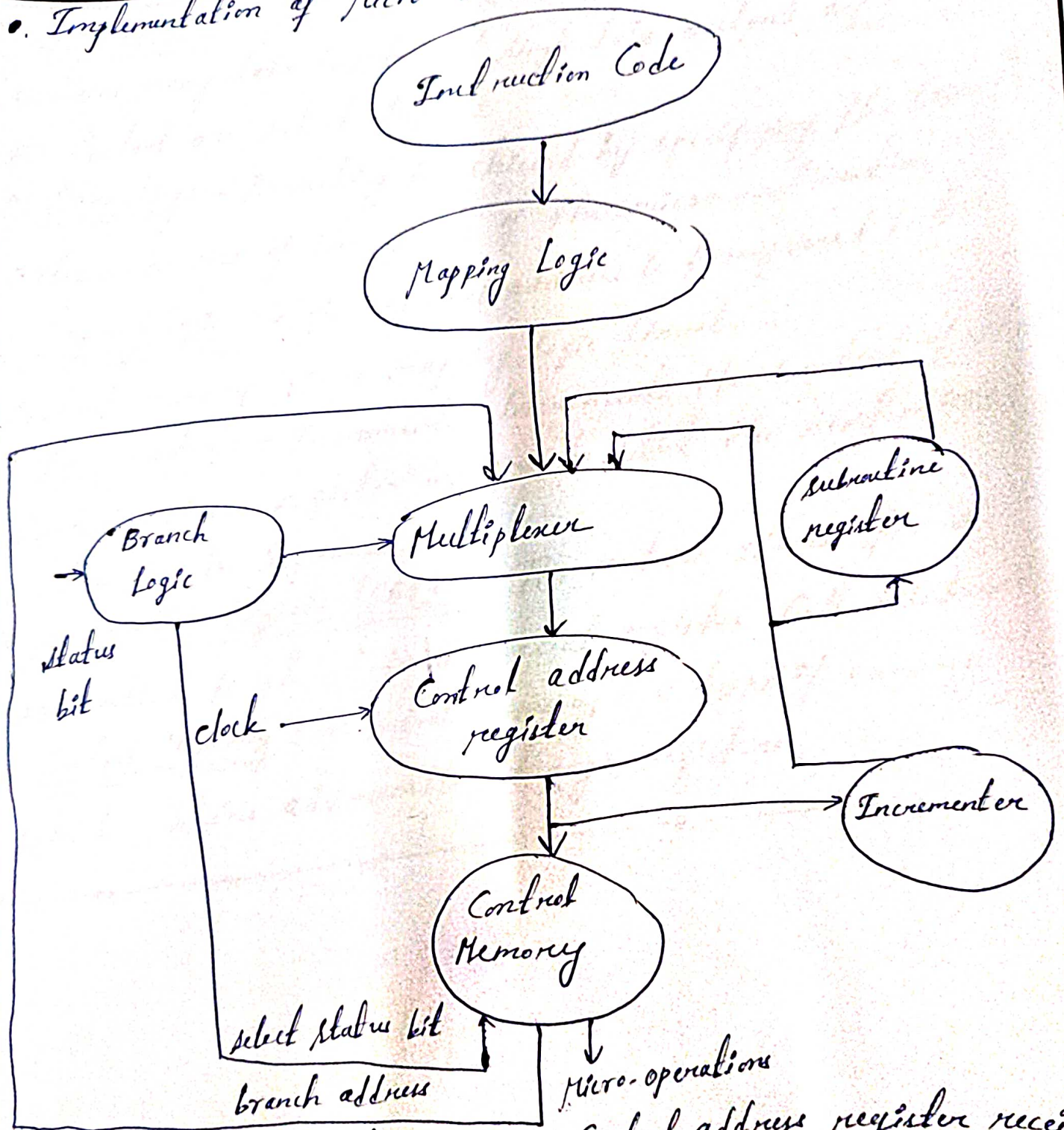
Q. 10 What is the purpose of a microprogram sequence? Explain its internal structure and work in detail.
 $\rightarrow$ Micro Instruction sequencer is a combination of all hardware for selecting the next micro-instruction address. The micro-instruction in control memory contains a set of bits to initiate micro-operations in computer registers and other bits to specify the method by which the address is obtained.



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• Implementation of Micro-Instructions Sequencer



• Control Address Register (CAR) → Control address register receives the address from four different paths.



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- Multiplexer $\rightarrow$ Multiplexer is a Combinational Circuit Which contains many data inputs and single data output depending on Control or select inputs.
 - Branching $\rightarrow$ Branching is achieved by specifying the branch address in one of the fields of the micro instruction.
 - Mapping logic $\rightarrow$ An external address is transferred into Control memory via a mapping logic circuit.
 - Incrementer $\rightarrow$ Incrementer increments the Content of the Control address register by one, to select the next micro-instruction in Sequence.
 - Subroutine Register (SBR) $\rightarrow$ The return address for a subroutine is stored in a Special register called subroutine.
 - Control Memory $\rightarrow$ Control memory is a type of memory which contains addressable storage registers.
-